
CORPUS-ADAPTIVE PIVOTAL ROUNDING FOR POOLED TEXT REPRESENTATIONS

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ABSTRACT

A compressed embedding table is reused across many texts, so correlated rounding errors can cancel after pooling even when entrywise errors do not decrease. We study unbiased adjacent-grid compression of a fixed table under nonnegative linear pooling. For any predictable pivotal rounding policy, an exact covariance decomposition expresses expected representation distortion through pairwise differences of token-frequency profiles and a terminal rounding term. One simultaneous event for finitely many corpus moments consequently certifies every policy selected from the calibration sample, including adaptively constructed candidates. The resulting comparison with the best policy has an explicit calibration remainder and needs no enumeration of the policy class. We develop tree-based evaluation and controlled optimality and lower-bound examples to clarify this guarantee. Reproducible experiments show substantial cancellation for uniform synthetic bags, but only modest improvements over independent rounding for a small corpus-derived static representation; biased nearest rounding performs better there. The results concern expected squared distortion of pooled vectors, not semantic quality or contextual language-model compression.

1 INTRODUCTION

Static text representations often begin with a table of token vectors and combine the retrieved rows. Averaging is particularly transparent: a document becomes a weighted sum of a shared table. This architecture appears in efficient text classification (Joulin et al., 2017), while widely used word-vector constructions provide examples of the tables one might compress (Mikolov et al., 2013; Pennington et al., 2014). Storing fewer bits per coordinate can reduce table storage, but the relevant approximation error depends on how the table is used. A coordinate error that is large for one token may cancel another token’s error in a pooled document.

This distinction matters because a compressed table is usually sampled or constructed once and then reused. Independent unbiased rounding treats each entry separately. It fixes the marginal error variance but does not exploit the corpus’s pooling weights. Dependent rounding can preserve the same marginal variances while changing the covariance between errors. The computational question is therefore not merely which grid to use, but which entries should share rounding decisions. The statistical question is whether a coupling selected using a finite corpus sample remains useful for future texts.

We address these questions for a deliberately restricted representation model: a fixed table, a uniform grid without clipping, and nonnegative normalized pooling weights. The table is fixed independently of the calibration sample. A policy uses that sample to choose pairwise pivotal rounding moves. We use established pivotal rounding and its preservation properties (Gandhi et al., 2006; Chauvet, 2012) to analyze corpus-dependent risk, evaluate supplied trees exactly, and characterize the information needed to choose their correlations.

The main technical obstacle is sample reuse. A pairing policy can depend on the same sample used to estimate its distortion, and can also adapt to intermediate rounding outcomes. A concentration inequality proved for one fixed quantizer does not by itself justify selecting

a quantizer from those estimates. We avoid a policy-by-policy union bound by expressing every policy’s risk as a nonnegative combination of the same finite collection of corpus moments. The coefficients vary with the policy, but their total mass is fixed by the table’s grid fractions. Concentrating the moments once controls every policy on the same event.

Theorem 1 gives this identity and a two-sided calibration certificate, including an explicit comparison for approximate empirical minimizers. Theorem 2 relates tree rounding to distances between token-frequency profiles, while Theorem 3 computes a supplied tree’s expected empirical risk without enumerating its random outcomes. Two controlled limits describe what this theory does and does not promise. Uniform independent token bags admit an exact optimum over all unbiased adjacent-grid couplings (Theorem 4). Rare two-token bags exhibit a calibration lower bound within the pivotal class despite a zero-risk informed oracle (Theorem 5).

The distinction between a vector statement and a text claim is essential. Our theorems apply to any distribution on the simplex. Text enters through token histograms and a shared static table; no linguistic structure is assumed or proved. The experiments use both distributions chosen to expose the mechanism and a small representation derived from AG News token counts. In the latter setting, selecting among five trees improves modestly on independent rounding, while nearest rounding is better. Thus the evidence supports a mathematical account of cancellation and calibration, rather than a general-purpose replacement for existing embedding-compression methods.

2 RELATED WORK

Compression of text representations. Methods for compressing word tables optimize different objects. Product quantization uses product codebooks for approximate vector reconstruction and search (Jégou et al., 2011). Compositional coding learns sums of codebook vectors (Shu & Nakayama, 2018); fastText.zip combines quantization with compression of text classifiers (Joulin et al., 2016). Other approaches include compact word-vector representations (Andrews, 2016), training with low-precision vectors (Lam, 2018), adaptive allocation of compression across embeddings (Kim et al., 2020), and online low-rank compression with semantic losses (Acharya et al., 2018). These designs can alter training, codebooks, or the objective itself. We hold the table and adjacent grid fixed and change only the joint law of unbiased rounding.

The relationship between vector approximation and task behavior also requires care. Yin & Shen (2018) analyze embedding dimensionality through a pairwise inner-product loss. May et al. (2019) study the downstream performance of compressed word embeddings using eigenspace overlap; their Theorem 3 bounds an overlap quantity under bounded entries and singular-value assumptions. Neither objective is the expected squared error of a shared rounded table under nonnegative pooling. Sentence-BERT produces sentence embeddings using contextual encoders and a different training construction (Reimers & Gurevych, 2019); our static lookup-and-pool model does not cover its nonlinear computation.

Dependent and correlated rounding. Marginal preservation, pairwise negative dependence, and floor-or-ceiling sums are classical properties of dependent rounding (Gandhi et al., 2006). The binary-tree constructions in that work are especially close to the tree policies in Section 5. Ordered pivotal sampling has an exact characterization and explicit second-order inclusion probabilities (Chauvet, 2012). Martingale methods for pivotal sampling (Chauvet, 2015) and exponential inequalities for sampling designs (Chauvet & Gerber, 2020) further delimit the established probabilistic machinery. We give self-contained finite calculations for the particular risk representation needed below rather than claiming a new dependent-rounding theorem.

Correlated quantization can improve distributed mean estimation by coordinating client randomness. Suresh et al. (2022) obtain bounds tied to variation among inputs and lower bounds for specified quantizer classes. The shared-table problem differs because the same rounded token rows occur with random, corpus-dependent weights in many future vectors. QSGD instead supplies an influential independent stochastic quantization baseline with

norm-based scaling (Alistarh et al., 2017). Vector-balancing methods such as the Gram-Schmidt walk provide substantially different sub-Gaussian discrepancy guarantees (Bansal et al., 2019); their vector-balancing objective differs from selecting pivotal policies for non-negative pooling.

The closest boundaries. DiscQuant uses calibration information to exploit low-discrepancy structure in quantization (Chee et al., 2025). Its spectral and moment-based analysis, including its partial-rounding step, is distinct from our fully rounded, nonnegative pooling model. Dependent leverage-score sampling also uses binary pivotal trees adapted to geometric information (Shimizu et al., 2024); its main guarantees concern active least squares rather than pooled embedding distortion. These constructions supply the established rounding and tree ingredients. Our analysis combines an exact edge-and-terminal risk identity, a single finite-moment event valid after policy selection, and an exact survivor-moment recursion for supplied trees. The controlled examples characterize the resulting calibration problem within its stated coupling classes.

3 PROBLEM SETUP

Table and corpus. Let $V, d \geq 1$ and fix $E \in \mathbb{R}^{V \times d}$. A text is represented by $a \in \mathcal{S}_V = \{a \geq 0 : \sum_i a_i = 1\}$, with pooled vector $a^\top E$. For normalized token counts, repeated occurrences contribute to the same coordinate of a . Let P be a distribution on $\mathcal{S}_V$. The calibration sample $a^{(1)}, \dots, a^{(m)}$ is iid from P , with $m \geq 1$, and is independent of the fixed table and a fresh evaluation text.

Fix a finite uniform grid of step $\Delta > 0$ that brackets every entry without clipping. Write $E_{ik} = \ell_{ik} + \Delta x_{ik}$, where ℓ_{ik} and $\ell_{ik} + \Delta$ are adjacent grid points and $x_{ik} \in [0, 1]$. At an endpoint either valid adjacent interval may be used with its integral fraction. A compressed table has $Q_{ik} = \ell_{ik} + \Delta B_{ik}$ with $B_{ik} \in \{0, 1\}$ and $\mathbb{E} B_{ik} = x_{ik}$. Unbiasedness is conditional on the calibration sample when the policy is selected using it. We measure

$$R_P(\pi) = \mathbb{E}_{a \sim P} \mathbb{E}_\pi \|a^\top (Q - E)\|_2^2, \quad \widehat{R}(\pi) = \frac{1}{m} \sum_{r=1}^m \mathbb{E}_\pi \|(a^{(r)})^\top (Q - E)\|_2^2. \quad (1)$$

These expectations average over the compressed table’s randomness. Deployment may reuse one realization; the theorem does not give a high-probability guarantee over that realization.

Pivotal policies. For each column, start at $y = x$. Choose two fractional coordinates i, j by any predictable rule: the choice may depend on calibration data and previous rounding outcomes but not future randomness or evaluation data. Set $\alpha = \min(1 - y_i, y_j)$ and $\beta = \min(y_i, 1 - y_j)$. Replace (y_i, y_j) by $(y_i + \alpha, y_j - \alpha)$ with probability $\beta/(\alpha + \beta)$, or by $(y_i - \beta, y_j + \beta)$ otherwise. At least one coordinate becomes integral. Repeat until at most one fractional coordinate remains, then round that survivor by an unbiased Bernoulli draw. Integral entries never move. All such columnwise policies form the pivotal class used below; dependence between columns is immaterial to squared Euclidean risk.

For column k , define $s_k = \sum_i x_{ik}$, $h_k = \{s_k\}(1 - \{s_k\})$, and $S_k = \sum_i x_{ik}(1 - x_{ik})$, where $\{s\} = s - \lfloor s \rfloor$. Set

$$C = \frac{\Delta^2}{2} \sum_{k=1}^d (S_k + h_k), \quad D_{ij} = \mathbb{E}_P (a_i - a_j)^2, \quad M_i = \mathbb{E}_P a_i^2.$$

Hats denote calibration averages. The corpus profile of token i is the function $z_i : a \mapsto a_i$ in $L^2(P)$, so $D_{ij} = \|z_i - z_j\|^2$. This is frequency-profile geometry, not distance between the rows of E .

4 CORPUS-ADAPTIVE RISK CERTIFICATES

A policy’s random sequence of pairings need not have a tractable distribution. Nevertheless, its entire contribution to risk has a small common set of statistical inputs. Let $w_{ij,k}$ be the

expected sum of $\alpha\beta$ over moves pairing i, j in column k , and let $u_{i,k}$ be the expected terminal Bernoulli variance assigned to survivor i . These expectations condition on the calibration sample and include the policy's internal randomness.

Theorem 1 (Risk identity and simultaneous calibration). *Under the table, iid calibration, and pivotal-policy assumptions of Section 3, w, u are nonnegative and, for every column,*

$$\text{Cov}(B_{\cdot k} - x_{\cdot k}) = \sum_{i < j} w_{ij,k} (e_i - e_j)(e_i - e_j)^\top + \text{diag}(u_{\cdot k}), \quad (2)$$

$$\sum_{j \neq i} w_{ij,k} + u_{i,k} = x_{ik}(1 - x_{ik}), \quad \sum_i u_{i,k} = h_k. \quad (3)$$

Consequently,

$$R_P(\pi) = \Delta^2 \sum_k \left(\sum_{i < j} w_{ij,k} D_{ij} + \sum_i u_{i,k} M_i \right) \leq R_{\text{ind}} := \Delta^2 \sum_{i,k} x_{ik}(1 - x_{ik}) M_i.$$

Let $L = V(V-1)/2 + V$ and $t = \log(2L/\eta)$, where $0 < \eta < 1$. With probability at least $1 - \eta$ over calibration, simultaneously for all pivotal policies, including those selected from that sample,

$$|R_P - \hat{R}| \leq \sqrt{\frac{2CR_P t}{m}} + \frac{Ct}{3m}, \quad \begin{cases} R_P \leq 2\hat{R} + \frac{8Ct}{3m}, \\ \hat{R} \leq 2R_P + \frac{8Ct}{3m}. \end{cases} \quad (4)$$

If $\hat{\pi}$ has empirical risk within $\varepsilon \geq 0$ of the infimum over a specified nonempty class Π , then on this same event

$$R_P(\hat{\pi}) \leq 4 \inf_{\pi \in \Pi} R_P(\pi) + \frac{8Ct}{m} + 2\varepsilon. \quad (5)$$

The class may be sample-dependent, in which case the infimum is over that realized class. A complete proof is in Appendix A.1.

Why the event survives policy selection. A pivot is a mean-zero increment with conditional covariance $\alpha\beta(e_i - e_j)(e_i - e_j)^\top$. Cross-time terms vanish by conditioning, yielding (2). Taking diagonal entries and the variance of the preserved total gives (3). The total coefficient mass is

$$\Delta^2 \sum_k \left(\sum_{i < j} w_{ij,k} + \sum_i u_{i,k} \right) = C.$$

Every statistic $(a_i - a_j)^2$ and a_i^2 lies in $[0, 1]$. A bounded Bernstein inequality therefore controls all L means on one event. Multiplying these inequalities by the nonnegative, possibly sample-dependent coefficients and applying Cauchy–Schwarz gives the first inequality in (4). No independence between the coefficients and the calibration sample is needed once this event holds. Applying $2\sqrt{xy} \leq x + y$ yields the displayed two-sided comparisons; applying them twice yields (5).

This argument identifies both an advantage and a limitation. Statistical complexity enters through L rather than the number of policies, but the factor C can be large and the constants are conservative. Equation (5) is not a claim that searching all pivotal policies is computationally feasible, nor that its additive term is small in a short calibration regime. The finite-tree construction below gives a computable empirical minimizer over supplied candidates.

Where nonnegativity matters. Off-diagonal covariances in (2) equal $-w_{ij,k}$. Since $a_i a_j \geq 0$, their contribution cannot increase risk relative to independent rounding with the same marginals. For signed pooling weights this conclusion need not hold. The concentration argument concerns bounded simplex statistics as stated; extending it to other representations requires new bounds and, potentially, a different comparison baseline.

5 TREE GEOMETRY AND EXACT RISK EVALUATION

A full binary tree on the V token indices provides a concrete pivotal policy. Each subtree is reduced to at most one fractional survivor; its two children are then combined. The topology can be chosen using calibration data. For a node U , let $\rho_U = \max_{i,j \in U} \|z_i - z_j\|_{L^2(P)}$.

Theorem 2 (Tree-profile bound). *For any such tree policy,*

$$R_P \leq \Delta^2 \left\{ \frac{d}{4} \sum_{U \text{ internal}} \rho_U^2 + \left(\sum_k h_k \right) \max_i \|z_i\|^2 \right\}. \quad (6)$$

If $V = 2^J$, the tree is complete, and the diameter at every depth- ℓ node is at most $H2^{-\ell/s}$, $s > 0$, its internal contribution is at most

$$\frac{d\Delta^2 H^2}{4} \sum_{\ell=0}^{J-1} 2^{\ell(1-2/s)}.$$

This sum is bounded independently of J for $s < 2$, equals J for $s = 2$, and is $O(V^{1-2/s})$ for fixed $s > 2$. The terminal term is retained in every case. See Appendix A.2.

Each internal node performs at most one pivot between surviving original leaves. Its conditional variance coefficient is at most $1/4$, and its paired profile distance is at most ρ_U . Summing these costs proves the first term of (6); the remaining survivor has total variance h_k . Counting 2^ℓ nodes gives the finite geometric series. The hypothesis describes a profile hierarchy. It neither asserts that text corpora have a particular dimension s nor follows from clustering the embedding rows.

Evaluating a supplied tree. For empirical risk use explicit profiles $z_i = (a_i^{(1)}, \dots, a_i^{(m)})/\sqrt{m} \in \mathbb{R}^m$. For one column, a subtree U has deterministic residual $r_U = \{\sum_{i \in U} x_i\}$. If $r_U > 0$, let I_U be its surviving original leaf and keep $\mu_U = \mathbb{E}z_{I_U}$, $\nu_U = \mathbb{E}\|z_{I_U}\|^2$, and expected internal cost A_U . If $r_U = 0$, survivor moments are dummy zeros. A fractional leaf has $(\mu, \nu, A) = (z_i, \|z_i\|^2, 0)$.

Theorem 3 (Exact survivor-moment recursion). *Fix a supplied full binary tree conditional on calibration, and use independent random coins at distinct nodes. If both children have nonzero residuals, put $q = r_L + r_R$ and*

$$v = \begin{cases} r_L r_R & q \leq 1, \\ (1 - r_L)(1 - r_R) & q > 1. \end{cases}$$

Then

$$A_U = A_L + A_R + v(\nu_L + \nu_R - 2\langle \mu_L, \mu_R \rangle). \quad (7)$$

For $q < 1$, mix both child moments with left weight $\lambda = r_L/q$; for $q > 1$, use $\lambda = (1 - r_L)/(2 - q)$. For $q = 1$ no survivor remains. If just one child has nonzero residual, pass its moments and add both costs; if neither does, only add costs. The exact empirical risk is

$$\hat{R} = \Delta^2 \sum_k (A_{\text{root},k} + h_k \nu_{\text{root},k}).$$

It takes $O(dmV)$ real-arithmetic operations and $O(mV)$ auxiliary storage, processing columns sequentially. Evaluating K supplied trees takes $O(KdmV)$ operations; their empirical minimizer has $\varepsilon = 0$ in (5). Appendix A.5 gives the complete recursion proof.

The survivor carries its original profile, not an averaged token vector. Independent child randomness factors the cross moment into $\langle \mu_L, \mu_R \rangle$; the scalar pivot probabilities depend only on the deterministic residuals. These two facts close the recursion without retaining exponentially many survivor configurations. The operation count excludes candidate-tree construction and assumes explicit profiles and exact real arithmetic. Our implementation uses tolerances at integral residuals and checks a small rational example, so numerical agreement is evidence about implementation rather than an exact-arithmetic guarantee.

6 OPTIMALITY AND CALIBRATION LOWER BOUNDS

An exactly solvable corpus model. Suppose a text consists of $n \geq 1$ independent uniform token draws from a vocabulary of size V , and a is its normalized histogram. This removes corpus geometry but exposes reuse of the same rounded row at multiple positions.

Theorem 4 (Uniform-bag optimum). *For this distribution and any fixed table satisfying Section 3,*

$$\inf_{\text{unbiased adjacent-grid couplings}} R_P = \Delta^2 \sum_k \left\{ \frac{S_k}{nV} + \left(1 - \frac{1}{n}\right) \frac{h_k}{V^2} \right\}. \quad (8)$$

Every pivotal policy attains this value. Independent rounding replaces h_k by S_k . For even V and all fractions equal to $1/2$, the independent-to-optimal risk ratio is $1 + (n-1)/V$. Appendix A.3 proves the result, including degenerate cases.

Under uniform independent token sampling, $Eaa^\top = I/(nV) + (1 - 1/n)\mathbf{1}\mathbf{1}^\top/V^2$. The diagonal term fixes the contribution of the marginal variances; only the variance of the rounded column sum remains to be minimized. An integer-valued random variable of mean s_k has variance at least h_k . Pivotal rounding attains it by producing only the two adjacent integer sums. Thus long bags can make shared-table correlation consequential even though no marginal error improves. Uniform bags are a controlled reference distribution, not a language model fitted to text.

A class-specific cost of learning correlations. There is also an information obstacle when useful pairs are rare. Fix $m \geq 1$, $1 \leq G \leq m+1$, $V = 3G+1$, $d = \Delta = 1$. Each of G triples has fractions $(1/2, 1/2, 1/2)$; the last token is an integral filler. For $\theta \in \{0, 1\}^G$, let P_θ place mass $p = 1/(m+1)$ on the bag $(e_{g1} + e_{g,2+\theta_g})/2$ in block g , and its remaining mass on the filler.

Theorem 5 (Rare-block calibration lower bound). *Every learner that uses m iid calibration bags to select a pivotal policy has some θ for which*

$$\mathbb{E}R_{P_\theta}(\hat{\pi}) \geq \frac{G}{16(m+1)} \left(\frac{m}{m+1} \right)^m \geq \frac{G}{48(m+1)}. \quad (9)$$

The expectation includes calibration, learner, and rounding randomness. For each fixed θ , an informed pivotal oracle has zero risk. The distribution family depends on m . See Appendix A.4.

Pairing the two tokens actually used by a block gives the oracle exact cancellation. When that block is unseen, its hidden bit stays uniform. The two possible cancellation edges share one row's variance budget, forcing average conditional loss at least $1/16$. A block is absent with probability $(m/(m+1))^m$. Summing blocks and averaging the hidden bits proves (9).

The restriction to pivotal policies is substantive. Within a triple, choosing $(B_1, B_2, B_3) = (1, 0, 0)$ or $(0, 1, 1)$ equally often cancels both possible bags, but has positive covariance between coordinates 2 and 3. Thus the same construction cannot lower-bound arbitrary unbiased couplings. Nor does (9) provide a high-probability lower bound or characterize one fixed natural corpus as m grows.

7 EMPIRICAL VALIDATION

We test three different questions: whether the predicted cancellation appears in a controlled model, whether missing calibration information produces the stated obstruction, and whether selecting correlations helps a concrete text-derived table. All studies use 20 seeds and retain individual results. Error bands report one standard error across seeds, not simultaneous confidence bands. Appendix B specifies preprocessing, randomization, exact-risk calculations, and resources.

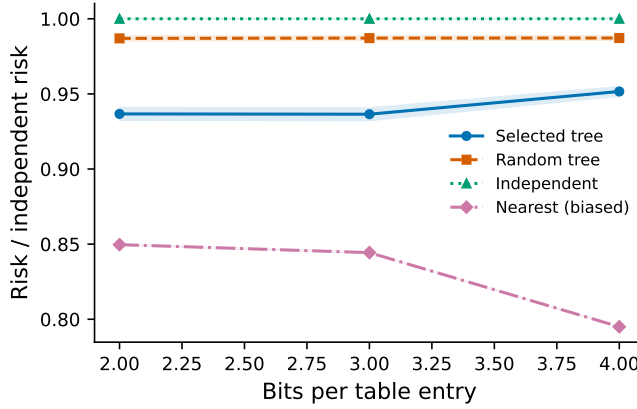


Figure 1: Expected pooled distortion on a fixed 2048-document AG News population. A 64-token, eight-dimensional spectral table is fitted on 5000 separate documents. With 128 iid calibration draws, select among four random pairing trees and one Ward-profile tree by exact expected empirical risk. Means and shaded standard errors use 20 seeds. Pivotal selection modestly reduces independent-rounding risk; biased nearest rounding performs better. The tree-risk recurrence and diameter bound are checked separately in the recorded experiment.

Corpus-derived static representation. We fit a 64-token, eight-dimensional spectral table on 5000 AG News documents, using the 63 most frequent training tokens and one other-token bucket. A separate set of 2048 documents defines a fixed empirical population of normalized histograms. Calibration consists of iid draws with replacement from that population, so the conditional experiment matches the sampling model in Theorem 1. The population is used in full for evaluation of expected distortion, not for choosing the policy. This finite-population design is distinct from evaluating transfer to unseen language.

For each $m \in \{32, 128, 512\}$ and bit budget $b \in \{2, 3, 4\}$, we compare independent rounding, biased nearest rounding, one random tree, a Ward tree built from calibration profiles, and empirical selection among four random trees plus the Ward tree. The grid spans the scaled table’s range $[-1, 1]$. Theorem 3’s recursion computes expected calibration and evaluation risk exactly in real arithmetic; numerical evaluations use floating point.

At $m = 128$, selected-tree risks divided by independent risks are 0.937, 0.936, and 0.952 at two, three, and four bits. The corresponding nearest-rounding ratios are 0.850, 0.844, and 0.795 (Figure 1). At $m = 512$, selection reaches ratios 0.927, 0.928, and 0.945. Independent absolute risks are 0.3581, 0.06550, and 0.01408. Thus increasing precision reduces absolute error far more than choosing among these five correlations.

Every one of the 180 selected-policy configurations satisfies the independent-rounding and diameter upper bounds within 10^{-10} . The diameter bound at $m = 128$ is about 2.82–3.57 times independent risk. The direct upper certificate $2\hat{R} + 8Ct/(3m)$ is looser still: its mean ratio to independent risk is about 20.4–20.7 at $m = 128$, falling to 6.50–6.55 at $m = 512$, with $\eta = 0.05$. These values limit its practical predictive usefulness on this population. Evaluating five supplied candidates takes roughly 0.011–0.017 seconds over these calibration sizes, excluding construction. A separate check samples 100 tables for each seed at three bits and $m = 128$; the largest standardized difference between the sampled mean and the exact expectation is 2.39 estimated standard errors.

Uniform bags. With $V = 64$, one coordinate, unit grid spacing, and midpoint fractions, fixed pairwise pivotal rounding has analytic risk $1/(4n)$. Independent rounding has risk $1/(4n) + (1 - 1/n)/256$. For each seed we generate 5000 bags and 32 shared compressed tables per method, at lengths 1, 4, 16, 64, 256, 1024. Figure 2 shows the continued decrease of pivotal error and the independent method’s plateau. At $n = 1024$ the exact risk ratio is 16.984375. The empirical means are 0.004451 and 0.0002447, with seed standard errors

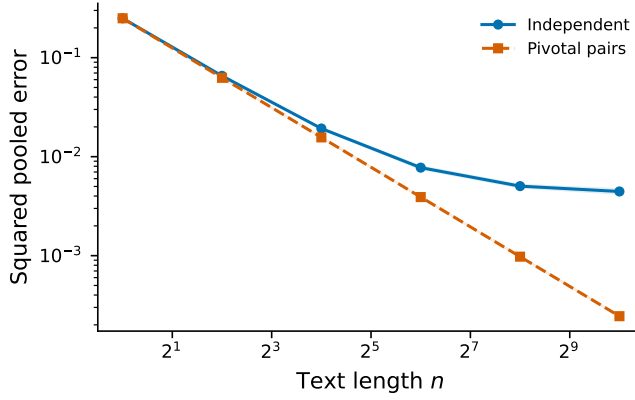


Figure 2: Shared-table cancellation for uniform iid token bags, $V=64$, $d=1$, unit grid spacing and midpoint fractions. Each of 20 seeds uses 5000 bags and 32 compressed tables per method; bands show one standard error across seeds. Pairwise pivotal rounding continues to improve with text length while independent rounding approaches a nonzero floor. This is a controlled exchangeable-token model.

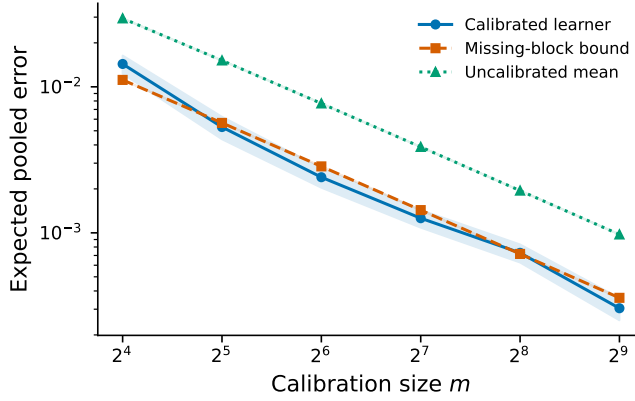


Figure 3: Rare-block calibration with $G=8$ disjoint midpoint triples and block probability $1/(m+1)$. The learner identifies observed co-occurrence bits and guesses unseen ones. The blue mean and band use 20 independent seeds and one standard error; other curves are analytic expectations. The population varies with m , as in the minimax construction. The missing-block lower bound applies to expected risk, so finite-seed means can lie below it; the oracle has zero risk.

0.0002280 and 0.0000001660, respectively. Finite-table sampling affects their ratio; the analytic comparison is the theorem's quantity.

Rare bags. For $G = 8$ and $m = 16, 32, 64, 128, 256, 512$, a learner identifies the bit of any observed block and guesses each unseen bit uniformly. We evaluate its population risk analytically given those guesses. This learner's expectation equals the displayed lower bound; a finite-seed mean need not exceed an expectation bound at every m . The mean at $m = 512$ is 0.0003046 ± 0.0000527 standard error, versus the lower bound 0.0003589 (Figure 3). This is not a violation: the bound averages over calibration and hidden bits, and the displayed mean uses only 20 seeds. The informed oracle is exactly zero.

What the validation establishes. The synthetic studies isolate mechanisms already specified mathematically. The corpus experiment checks their implementation on nonuniform text-derived weights and measures the actual size of the available gains. Its small

vocabulary and dominant other-token bucket restrict generality. We do not measure retrieval, classification, semantic similarity, perplexity, or hardware throughput. In particular, nearest rounding’s better distortion is a useful negative result for a broad superiority claim; unbiasedness is a modeling constraint, not evidence that this constraint is optimal for an application.

8 CONCLUSION AND LIMITATIONS

For a fixed table under nonnegative pooling, pivotal rounding turns corpus geometry into a choice of error correlations. Its expected distortion admits a nonnegative edge-and-terminal decomposition. That identity supports a calibration event valid after policy selection, an exact recursion for supplied trees, and controlled examples showing both complete cancellation and missing-information costs. The results explain a restricted compression problem without changing marginal quantization errors.

The principal limitations are operational. The calibration certificate can be loose, tree construction is not optimized by the risk recursion, and expected distortion does not certify a single deployed table. The optimality example uses exchangeable uniform tokens, while the lower bound uses a sample-size-dependent rare-block family and only the pivotal comparator. The text experiment uses a small corpus-derived static representation and favors nearest rounding in absolute distortion. Extending the analysis to signed or nonlinear pooling, proving useful one-table guarantees, and identifying efficiently searchable couplings beyond pivotal trees remain concrete questions.

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APPENDIX

A DETAILED PROOFS

All risks below condition on the observed calibration sample when a policy uses it. The only outer probability in the calibration certificate concerns that sample. Rounding randomness is integrated in the definition of each risk. We keep these two sources of randomness separate throughout.

A.1 COVARIANCE DECOMPOSITION AND CALIBRATION

Proof of Theorem 1. Step 1: one pivotal increment. Fix a column and suppress its index. The table and grid fix $x \in [0, 1]^V$. Conditional on a calibration sample, expose all policy randomization through a filtration $(\mathcal{F}_t)$. If the rule randomizes the next pair, include that choice in the history just before its pivot coin. Thus the selected pair and its current values are measurable before the coin is drawn. Set $v_{ij} = e_i - e_j$. A pivot increment ξ_t equals αv_{ij} with probability $\beta/(\alpha + \beta)$ and $-\beta v_{ij}$ with probability $\alpha/(\alpha + \beta)$. Both α, β are positive because both entries are fractional. Direct calculation gives

$$\mathbb{E}(\xi_t \mid \mathcal{F}_t) = \frac{\alpha\beta - \beta\alpha}{\alpha + \beta} v_{ij} = 0, \quad (10)$$

$$\mathbb{E}(\xi_t \xi_t^\top \mid \mathcal{F}_t) = \frac{\alpha^2\beta + \beta^2\alpha}{\alpha + \beta} v_{ij} v_{ij}^\top = \alpha\beta v_{ij} v_{ij}^\top. \quad (11)$$

A terminal survivor of value r at index i has increment $(H - r)e_i$, where H is Bernoulli with parameter r ; its conditional second moment is $r(1 - r)e_i e_i^\top$ and its conditional mean is zero.

Each pivot fixes at least one fractional entry, so there are at most $V - 1$ pivots, followed by at most one terminal draw. Pad the process with zero increments to a deterministic finite horizon. This avoids any stopping-time integrability issue. Every increment is bounded. For $t < q$, ξ_t is measurable before the later coin, and

$$\mathbb{E}[\xi_t \xi_q^\top] = \mathbb{E}[\xi_t \mathbb{E}(\xi_q^\top \mid \mathcal{F}_q)] = 0.$$

The transposed cross term vanishes as well. Summing the increments establishes $\mathbb{E}B = x$ and shows that the covariance of $B - x$ is the sum of the expected conditional second moments. Define w_{ij} as the expected accumulated coefficient $\alpha\beta$ for pair i, j , and u_i as the expected terminal variance at i . Equation (11) proves (2). In particular $w, u \geq 0$, and for distinct coordinates the covariance is $-w_{ij}$.

Step 2: variance and total-mass bookkeeping. The marginal B_i is binary with mean x_i , hence $\text{Var}(B_i) = x_i(1 - x_i)$. Taking diagonal entries of the covariance identity yields

$$x_i(1 - x_i) = \sum_{j \neq i} w_{ij} + u_i. \quad (12)$$

All pivots preserve the column sum $s = \sum_i x_i$. Just before the terminal draw the integral entries have some integer sum and the sole possible residual equals $\{s\}$. Thus $\sum_i B_i$ is $\lfloor s \rfloor$ or $\lceil s \rceil$, with variance $h = \{s\}(1 - \{s\})$. If s is integral there is no fractional survivor and this variance is zero. Multiplying the covariance identity by $\mathbf{1}$ on both sides annihilates its edge terms, so

$$\sum_i u_i = h, \quad S = 2 \sum_{i < j} w_{ij} + \sum_i u_i, \quad \sum_{i < j} w_{ij} + \sum_i u_i = \frac{S + h}{2}. \quad (13)$$

The terminal mass cannot be discarded when s is not an integer. These identities also show that the total mass is determined by the table, not by which policy or calibration sample selected the edges.

Step 3: the risk identity and independent comparator. For a fixed text a , unbiasedness gives

$$\mathbb{E}_\pi \left[(a^\top (B - x))^2 \right] = \sum_{i < j} w_{ij} (a_i - a_j)^2 + \sum_i u_i a_i^2.$$

Multiply by Δ^2 , sum over columns, and average over P to obtain the displayed identity in Theorem 1. There are no cross-column products because squared Euclidean norm is the sum of coordinate squares. Expanding the covariance instead by its diagonal and off-diagonal entries gives

$$\mathbb{E}_\pi \left[(a^\top (B - x))^2 \right] = \sum_i x_i (1 - x_i) a_i^2 - 2 \sum_{i < j} w_{ij} a_i a_j.$$

Nonnegative pooling makes the second term nonpositive. The first is exactly the independent-rounding variance. Averaging proves the independent comparison. The use of the simplex in this step is its nonnegativity; its normalization will also bound the scalar statistics below.

Auxiliary bounded-moment inequality. We prove the concentration inequality with its constants. Let $Z \in [0, 1]$ have mean μ and let $Y = Z - \mu$. Then $\mathbb{E}Y = 0$, $|Y| \leq 1$, and $\mathbb{E}Y^2 \leq \mathbb{E}Z^2 \leq \mu$. For each integer $r \geq 2$, $\mathbb{E}|Y|^r \leq \mu$. Also $r! \geq 2 \cdot 3^{r-2}$: it holds at $r = 2$, and multiplying by $r + 1 \geq 3$ propagates it. For either sign and $0 \leq \lambda < 3$, the absolutely convergent exponential series therefore implies

$$\mathbb{E}e^{\pm \lambda Y} \leq 1 + \mu \sum_{r=2}^{\infty} \frac{\lambda^r}{r!} \leq 1 + \frac{\mu \lambda^2}{2(1 - \lambda/3)}, \quad (14)$$

$$\log \mathbb{E}e^{\pm \lambda Y} \leq \frac{\mu \lambda^2}{2(1 - \lambda/3)}. \quad (15)$$

The last step uses $\log(1 + u) \leq u$ for $u \geq 0$. For m independent copies, independence adds their logarithmic moment bounds, giving variance proxy $v = m\mu$.

If $\mu = 0$, nonnegativity implies $Z = 0$ almost surely: for every positive integer j , Markov's inequality gives $\mathbb{P}(Z \geq 1/j) = 0$, and their countable union contains $\{Z > 0\}$. Suppose now $v > 0$ and let $t > 0$. Write $c = 1/3$, $q = \sqrt{2t/v}$, $\lambda = q/(1 + cq)$, and $b = \sqrt{2vt} + ct = vq + cvq^2/2$. Then $0 < \lambda < 3$ and

$$\lambda b - \frac{v\lambda^2}{2(1 - c\lambda)} = \frac{vq^2(1 + cq/2)}{1 + cq} - \frac{vq^2}{2(1 + cq)} = \frac{vq^2}{2} = t. \quad (16)$$

Exponential Markov's inequality applied to each sign in (15) gives the two-sided bound

$$\mathbb{P} \left(|\bar{Z} - \mu| > \sqrt{\frac{2\mu t}{m}} + \frac{t}{3m} \right) \leq 2e^{-t}. \quad (17)$$

This derivation uses no unproved concentration theorem.

Step 4: a single event for every policy. The $L = V(V - 1)/2 + V$ fixed statistics are $Z_{ij}(a) = (a_i - a_j)^2$, $i < j$, and $Z_i(a) = a_i^2$. They belong to $[0, 1]$ on $\mathcal{S}_V$. Their true means are D_{ij}, M_i ; their empirical means use the iid calibration sample. Apply (17) to each statistic and take a union bound with $t = \log(2L/\eta)$. With probability at least $1 - \eta$, all scalar inequalities hold together. Dependence between the statistics on a document does not affect this union bound.

Fix any sample in this event. Only now select a pivotal policy using that sample. Aggregate the nonnegative coefficients across columns: for an edge use $c_{ij} = \Delta^2 \sum_k w_{ij,k}$, and for a node use $c_i = \Delta^2 \sum_k u_{i,k}$. Index this collection by l , and denote the matching true and empirical means by $\mu_l, \hat{\mu}_l$. Equation (13) gives

$$c_l \geq 0, \quad \sum_l c_l = C, \quad R_P = \sum_l c_l \mu_l, \quad \hat{R} = \sum_l c_l \hat{\mu}_l.$$

Consequently the scalar bounds and Cauchy–Schwarz give

$$|\hat{R} - R_P| \leq \sqrt{\frac{2t}{m}} \sum_l c_l \sqrt{\mu_l} + \frac{Ct}{3m}, \quad (18)$$

$$\left(\sum_l c_l \sqrt{\mu_l} \right)^2 = \left(\sum_l \sqrt{c_l} \sqrt{c_l \mu_l} \right)^2 \leq CR_P. \quad (19)$$

This proves the first inequality in (4). It is a deterministic implication for every such coefficient vector, so the event remains valid for all sample-selected policies and all realized sample-selected classes. No union bound over a potentially infinite policy space is taken. The same conditional policy law must be used in $\hat{R}$ and R_P , as in (1).

Step 5: constants and the comparison of infima. Put $b = Ct/m$. The inequality $\sqrt{2bR_P} \leq R_P/2 + b$ follows from $(\sqrt{R_P} - \sqrt{2b})^2 \geq 0$. Hence

$$R_P \leq \hat{R} + \frac{R_P}{2} + \frac{4b}{3}, \quad R_P \leq 2\hat{R} + \frac{8b}{3}.$$

For the other direction, $\sqrt{2bR_P} \leq R_P + b/2$ implies

$$\hat{R} \leq 2R_P + \frac{5b}{6} \leq 2R_P + \frac{8b}{3}.$$

The symmetric constant in the theorem is conservative. For an empirical ε -minimizer over a nonempty class, apply the first comparison to $\hat{\pi}$, use its defining empirical inequality, then apply the second comparison to each competitor:

$$R_P(\hat{\pi}) \leq 2 \inf_{\pi \in \Pi} \hat{R}(\pi) + 2\varepsilon + \frac{8b}{3} \quad (20)$$

$$\leq 4 \inf_{\pi \in \Pi} R_P(\pi) + 2\varepsilon + \frac{16b}{3} + \frac{8b}{3}. \quad (21)$$

This is (5). Taking the infimum of pointwise inequalities requires neither attainment nor interchange of expectation and optimization. If an exact minimizer is unavailable, the assertion applies whenever the stated approximate minimizer exists.

Boundary checks and conclusion. When $C = 0$, all coefficients vanish and both risks are zero. A zero population moment is zero empirically almost surely by the nonnegativity argument above. Integral fractions never move, and $V = 1$ leaves only the terminal statistic, which is identically one for simplex-valued texts. Since $m \geq 1$, $L \geq 1$, and $0 < \eta < 1$, all displayed denominators and logarithms are well-defined. The table’s independence from calibration fixes the fractions and their mass; the iid hypothesis is used in (17); predictability is used in (11). Thus the covariance identity, independent comparison, simultaneous event, and oracle inequality all hold under exactly the stated assumptions, proving Theorem 1. $\square$

A.2 PROFILE GEOMETRY ALONG A TREE

Proof of Theorem 2. Fix a column and a full binary tree. We first justify the reduction at each node. A leaf is either integral or has one fractional coordinate. Inductively, suppose each child subtree has at most one fractional survivor. If both have survivors, one pivot between them makes at least one integral, leaving at most one for their parent. If only one has a survivor, no pivot is needed. If neither has one, the parent has none. This induction establishes both the algorithm’s validity and the fact that a survivor is always an original leaf index. The tree does not replace original profiles by their averages.

Step 1: bound a node’s conditional cost. If an internal node U pivots indices i, j , both are leaves of U and lie in disjoint child sets. The risk identity (2), or directly (11), shows that the conditional contribution to expected squared pooled error, divided by Δ^2 , is $\alpha\beta\|z_i - z_j\|^2$. The pair may be random because the survivors depend on earlier coins, but $\|z_i - z_j\| \leq \rho_U$ for every outcome.

To verify the scalar constant, write the two current fractions as $a, b \in (0, 1)$. If $a + b \leq 1$, then $\alpha = b$ and $\beta = a$, so $\alpha\beta = ab \leq (a + b)^2/4 \leq 1/4$. If $a + b \geq 1$, then $\alpha = 1 - a$, $\beta = 1 - b$, and

$$\alpha\beta = (1 - a)(1 - b) \leq \frac{(2 - a - b)^2}{4} \leq \frac{1}{4}.$$

Equality at $a + b = 1$ is consistent with both formulas. Thus, whether the node pivots or simply passes a survivor, its conditional expected cost is at most $\rho_U^2/4$. Taking expectation preserves this bound.

Step 2: add internal and terminal terms. There is at most one pivot at each internal node. Cross-time martingale terms vanish by the proof of Theorem 1; one must add variances rather than square a sum of their upper bounds. Consequently, for a column,

$$\mathbb{E} \sum_{\text{pivots at } U} \alpha\beta \|z_i - z_j\|^2 \leq \frac{1}{4} \sum_{U \text{ internal}} \rho_U^2.$$

At the root, the remaining fractional mass is $\{s_k\}$. If its original index is I , its conditional Bernoulli variance contribution is $h_k \|z_I\|^2$. Therefore it is bounded by $h_k \max_i \|z_i\|^2$. This term is zero exactly when $h_k = 0$; zero profile diameters alone do not eliminate it. Summing the internal bound over d columns and the residual bound over k and restoring Δ^2 proves (6).

The proof allows a tree to be chosen from calibration data, because the inequality is conditional on that choice and concerns the population profile distances of the resulting tree. No assumption about a construction algorithm was used. Conversely, computing a small empirical diameter is not itself a population-diameter guarantee; that is a separate statistical question.

Step 3: evaluate the hierarchy. For a complete tree with $V = 2^J$ leaves there are 2^ℓ internal nodes at depth ℓ , for $\ell = 0, \dots, J - 1$. The hypothesis $\rho_U \leq H2^{-\ell/s}$ gives

$$\sum_{U \text{ internal}} \rho_U^2 \leq H^2 \sum_{\ell=0}^{J-1} 2^\ell 2^{-2\ell/s} = H^2 \sum_{\ell=0}^{J-1} q^\ell, \quad q = 2^{1-2/s}.$$

For $s < 2$, $q < 1$ and the sum is $(1 - q^J)/(1 - q) \leq 1/(1 - q)$. For $s = 2$, $q = 1$ and the sum consists of J ones. For $s > 2$, $q > 1$ and

$$\sum_{\ell=0}^{J-1} q^\ell = \frac{q^J - 1}{q - 1} = \frac{V^{1-2/s} - 1}{2^{1-2/s} - 1}.$$

These are finite algebraic identities; the constant in the last order statement depends on fixed s and diverges as $s \downarrow 2$. The separate middle case is therefore necessary.

Boundary checks and conclusion. If $V = 1$ then $J = 0$, there are no internal nodes, and the terminal bound alone applies. If every diameter is zero, every pair increment cancels on every profile, but a fractional total may still incur root error. If the table is integral, both internal and terminal costs vanish. The proof uses only the adjacent-grid variance scale, the pivotal update, and the one-survivor property of a binary tree. It neither assumes nor establishes a language-specific dimensional law. Combining the node bound, residual calculation, and finite series proves every part of Theorem 2. $\square$

A.3 THE UNIFORM-BAG OPTIMUM

Proof of Theorem 4. Let $T_1, \dots, T_n$ be iid uniform indices in $\{1, \dots, V\}$ and $a = n^{-1} \sum_{r=1}^n e_{T_r}$. Rounding is independent of these fresh evaluation draws, conditional on the chosen coupling. The argument permits any unbiased adjacent-grid coupling, not only a pivotal one.

Step 1: calculate the second moment of pooling. Expanding the outer product separates equal and unequal token positions:

$$\mathbb{E}aa^\top = \frac{1}{n^2} \sum_{r=1}^n \mathbb{E}e_{T_r}e_{T_r}^\top + \frac{1}{n^2} \sum_{r \neq q} \mathbb{E}e_{T_r}e_{T_q}^\top \quad (22)$$

$$= \frac{I}{nV} + \frac{n(n-1)}{n^2V^2} \mathbf{1}\mathbf{1}^\top. \quad (23)$$

For $r = q$, the expected coordinate projector is I/V . For $r \neq q$, independence gives the product of uniform means $\mathbf{1}\mathbf{1}^\top/V^2$. This calculation includes repeated draws of the same token and is the reason a column-sum variance term survives even for long bags.

Fix a column and put $X = B - x$ and $\Sigma = \mathbb{E}XX^\top$. Since $\mathbb{E}X = 0$, Σ is its covariance. The diagonal entries are fixed by the Bernoulli marginals, so $\text{tr } \Sigma = S$. Using (23),

$$\mathbb{E}(a^\top X)^2 = \text{tr}(\Sigma \mathbb{E}aa^\top) \quad (24)$$

$$= \frac{S}{nV} + \left(1 - \frac{1}{n}\right) \frac{\mathbf{1}^\top \Sigma \mathbf{1}}{V^2} = \frac{S}{nV} + \left(1 - \frac{1}{n}\right) \frac{\text{Var } K}{V^2}, \quad (25)$$

where $K = \sum_i B_i$ is integer-valued and $\mathbb{E}K = s = \sum_i x_i$. There are no other degrees of freedom in this objective.

Step 2: minimize the integer-sum variance. Write $a_0 = \lfloor s \rfloor$ and $r = s - a_0 \in [0, 1)$. For every integer K , the product $(K - a_0)(K - a_0 - 1)$ is nonnegative: the two factors are consecutive integers, so they cannot have opposite nonzero signs. Taking expectation and expanding around the mean gives

$$0 \leq \mathbb{E}[(K - a_0)(K - a_0 - 1)] \quad (26)$$

$$= \mathbb{E}K^2 - (2a_0 + 1)s + a_0(a_0 + 1) \quad (27)$$

$$= \text{Var } K + s^2 - (2a_0 + 1)s + a_0(a_0 + 1) \quad (28)$$

$$= \text{Var } K - r(1 - r). \quad (29)$$

Thus $\text{Var } K \geq r(1 - r) = h$. The coefficient multiplying this variance in (25) is nonnegative for $n \geq 1$, so substituting h gives a valid lower bound for every coupling. At $n = 1$ that coefficient is zero, which makes the bound independent of the sum law without invalidating it.

Step 3: attain the bound jointly with all marginals. A pivotal policy preserves the real sum during every pair move and leaves at most one fraction. The final Bernoulli draw therefore produces $K = a_0$ or $a_0 + 1$, with probabilities $1 - r$ and r . Its sum variance is exactly h , and Step 1 of the certificate proof established every required marginal. Hence a feasible coupling simultaneously has the prescribed marginals and the minimum sum variance. Substitution in (25), multiplication by Δ^2 , and addition over columns prove (8). Arbitrary dependence across columns cannot lower the result because the objective is the sum of their separate squared errors, and choosing pivotal couplings columnwise attains all lower bounds together.

For independent rounding, $\text{Var } K = \sum_i \text{Var } B_i = S$. Replacing h by S in the optimum formula therefore gives its exact risk. This comparison uses identical grid fractions and differs only in the joint law.

Step 4: midpoint ratio and special cases. If V is even and every fraction is $1/2$, then $S_k = V/4$ and $s_k = V/2$ is integral, so $h_k = 0$. With $d \geq 1$ and $\Delta > 0$,

$$R_* = \frac{d\Delta^2}{4n}, \quad R_{\text{ind}} = \frac{d\Delta^2}{4n} + \left(1 - \frac{1}{n}\right) \frac{d\Delta^2}{4V}.$$

The denominator is positive and division gives $R_{\text{ind}}/R_* = 1 + (n-1)/V$. At $V = 64, n = 1024$ this equals $1 + 1023/64 = 16.984375$.

For an integral table, $S_k = h_k = 0$ in every column and both risks vanish; no ratio is asserted. If $V = 1$, then $h_k = S_k = x_{1k}(1 - x_{1k})$, and the two terms in (8) add to $\Delta^2 S_k$, as required because every text uses the sole token. At $n = 1$, both risks coincide with $\Delta^2 \sum_k S_k / V$. These checks also show why a positive-risk condition is needed for any general ratio. The exact moment identity, integer lower bound, and pivotal attainment together prove Theorem 4. $\square$

A.4 RARE BLOCKS AND UNOBSERVED PAIR IDENTITIES

Proof of Theorem 5. Fix the integers m, G in the theorem. There are G disjoint triples and one filler coordinate, so $V = 3G + 1$. Use the grid $\{0, 1\}$, unit spacing, midpoint entries within each triple, and the zero entry for the filler. Let $p = 1/(m + 1)$. The specified masses form a probability distribution because $Gp \leq 1$; the filler mass may be zero when $G = m + 1$. Each two-token bag lies in the simplex and uses weights $1/2, 1/2$.

Step 1: construct an informed zero-risk policy. Suppose the bit vector θ is known. In each triple, first pivot the two coordinates $g1$ and $g, 2 + \theta_g$ that appear in its bag. They both begin at $1/2$, so their two outcomes are $(1, 0)$ and $(0, 1)$, each with probability $1/2$. Their original sum and rounded sum both equal one. Hence

$$\frac{1}{2}(B_{g1} - 1/2) + \frac{1}{2}(B_{g,2+\theta_g} - 1/2) = 0$$

on every rounding outcome. Those two coordinates are integral thereafter. Process all remaining fractional coordinates by any legal pivotal rule, ending with an unbiased terminal draw if necessary. Those later moves do not alter any bag's already-fixed active pair. The filler is integral from the start and incurs no error. This is a feasible pivotal policy and its population risk is exactly zero for every fixed θ .

Step 2: the loss of an arbitrary selected pivotal policy. For any policy selected from a calibration sample, condition on that sample and on any learner randomness used to choose its law. The covariance identity applies even if future pair choices adapt to rounding outcomes or cross block boundaries. Within one triple, each marginal variance is $1/4$. For $j = 2$ or 3 , the expected squared error of its possible bag is therefore

$$L_{1j} = \mathbb{E} \left[\left(\frac{B_{g1} - 1/2 + B_{gj} - 1/2}{2} \right)^2 \right] \quad (30)$$

$$= \frac{1}{4} \left(\frac{1}{4} + \frac{1}{4} + 2 \text{Cov}(B_{g1}, B_{gj}) \right) = \frac{1}{8} - \frac{1}{2} w_{g1,gj}. \quad (31)$$

The coefficient w is the full expected accumulated pivotal edge weight, not just the probability of pairing the two coordinates first. The diagonal budget in (3) implies

$$w_{g1,g2} + w_{g1,g3} \leq \sum_{j \neq g1} w_{g1,j} + u_{g1} = \frac{1}{4}. \quad (32)$$

All omitted terms are nonnegative. Thus cross-block pairings cannot evade this budget.

Step 3: condition on a missing block. For a lower bound, temporarily draw the bits $\theta_1, \dots, \theta_G$ independently and uniformly. Let A_g be the event that no calibration draw falls in block g . Each draw chooses block g with probability p , regardless of θ . Hence

$$\mathbb{P}(A_g) = (1 - p)^m.$$

Conditional on A_g , the entire observed sample contains no bag whose value depends on θ_g . The other bits are independent of it, and the block-selection probabilities are fixed. Therefore θ_g remains uniform conditional on the sample and A_g . A learner's additional randomness, being independent of the unknown bits given its observed data, supplies no information about it.

For any realized such sample, the selected policy law consequently has the same weights whether $\theta_g = 0$ or 1. Averaging (31) over these two cases and using (32) gives

$$\mathbb{E}_{\theta_g}[L_{1,2+\theta_g} \mid A_g, \text{sample}] = \frac{1}{8} - \frac{1}{4}(w_{g1,g2} + w_{g1,g3}) \quad (33)$$

$$\geq \frac{1}{8} - \frac{1}{16} = \frac{1}{16}. \quad (34)$$

The same bound holds after averaging learner randomness. It does not assert that both possible bags have loss at least $1/16$; the average over the unknown bit is essential.

Step 4: sum losses and remove the prior. For any sample and bit vector, each block's contribution to population risk is its probability p times its bag loss. Losses are nonnegative. Retaining only outcomes with an unseen block and applying (34) yields

$$\mathbb{E}_{\theta, \text{sample}, \text{learner}} R_{P_\theta}(\hat{\pi}) \geq \sum_{g=1}^G p \mathbb{P}(A_g) \mathbb{E}[L_{1,2+\theta_g} \mid A_g] \quad (35)$$

$$\geq \frac{Gp}{16} (1-p)^m = \frac{G}{16(m+1)} \left(\frac{m}{m+1} \right)^m. \quad (36)$$

We have already integrated rounding randomness in each L . The events A_g need not be independent: linearity of expectation is sufficient. Since the prior is a uniform average over the finite set of bit vectors, at least one fixed θ has expected sample-and-learner risk at least that average. This establishes the first bound with the theorem's order of quantifiers, namely every learner has some difficult fixed bit vector.

To obtain the simpler constant, the binomial theorem and $\binom{m}{j}/m^j \leq 1/j!$ give

$$\left(1 + \frac{1}{m}\right)^m = \sum_{j=0}^m \binom{m}{j} m^{-j} \leq \sum_{j=0}^m \frac{1}{j!} < 3.$$

For the last inequality, $j! \geq 2^{j-1}$ for $j \geq 1$, with strict inequality for some j , so the infinite reciprocal-factorial sum is less than $1 + 1 + \sum_{j=2}^{\infty} 2^{1-j} = 3$. Taking reciprocals shows $(m/(m+1))^m > 1/3$; the weak $1/3$ suffices in (36). This proves the second bound in (9).

Step 5: boundaries of the comparator and construction. At $m = 1$, the missing-block factor equals $1/2$ and is well-defined. At $G = m + 1$, the filler mass is zero, but every block can still be absent from a finite sample with probability $(1-p)^m > 0$. The argument allows randomized learners, arbitrary predictable pivotal moves, and rounding across blocks; all share the nonnegative row budget. It is an expectation bound and does not imply a bound for each calibration sample.

For comparison, an unrestricted unbiased law within one triple may choose $(1, 0, 0)$ and $(0, 1, 1)$ each with probability $1/2$. Every coordinate has mean $1/2$, and both sums $B_1 + B_2$ and $B_1 + B_3$ equal one. Both possible bags thus have zero error without knowing the bit. But $\text{Cov}(B_2, B_3) = 1/4 > 0$, which is impossible under (2). Applying this law independently to all triples evades the lower bound outside the pivotal class. The theorem deliberately retains that restriction.

Finally, the probability p changes with m , so the construction is a family of distributions indexed by sample size; it does not describe convergence on one fixed text corpus. The zero-risk oracle, conditional unseen-bit loss, and finite-prior argument prove precisely Theorem 5. $\square$

A.5 EXACT EVALUATION BY SURVIVOR MOMENTS

Proof of Theorem 3. Condition on the calibration sample and the supplied tree topology. Its V leaves carry the explicit vectors $z_i \in \mathbb{R}^m$ with the normalization in Section 5. Treat one embedding column first. Random coins at different nodes are independent. A subtree's internal randomness therefore depends only on its own coins, which are disjoint from those

of its sibling. This assumption is stronger than the predictability sufficient for Theorem 1 and is used below to factor one cross moment.

Step 1: deterministic residuals and state variables. The subtree induction in the geometry proof leaves at most one fractional coordinate. Every pivot within a subtree preserves its coordinate sum. If the sum of the original fractions in U is s_U , all but at most one current coordinate are integers, so the residual must be $r_U = \{s_U\}$. It is deterministic conditional on the table, regardless of which leaf survives. When $r_U > 0$, define the random original index I_U and moments

$$\mu_U = \mathbb{E}z_{I_U}, \quad \nu_U = \mathbb{E}\|z_{I_U}\|^2.$$

Let A_U be the expectation of the sum of all internal pivot costs $\alpha\beta\|z_i - z_j\|^2$ in U . This quantity excludes a terminal Bernoulli rounding of its survivor: such rounding occurs only once, at the entire tree's root. Rounding every subtree's survivor separately would implement a different law.

For a leaf with $x_i \in (0, 1)$, $r_U = x_i$, $\mu_U = z_i$, $\nu_U = \|z_i\|^2$, and $A_U = 0$. For an integral leaf, $r_U = 0$ and all three state values can be set to zero. Dummy moments at zero residuals are never interpreted as expectations of an existing survivor.

Step 2: residual sums below one. Suppose both children have residuals $a = r_L > 0$ and $b = r_R > 0$, with $q = a + b < 1$. The pivot has $\alpha = b$, $\beta = a$. With probability a/q its outcome on the surviving indices is $(q, 0)$; with probability b/q it is $(0, q)$. Thus the parent survivor is the left original leaf with probability $\lambda = a/q$ and the right original leaf otherwise. These probabilities depend only on a, b , not on which leaves survived. Therefore

$$r_U = q, \quad \mu_U = \frac{a}{q}\mu_L + \frac{b}{q}\mu_R, \quad \nu_U = \frac{a}{q}\nu_L + \frac{b}{q}\nu_R. \quad (37)$$

The pivot variance coefficient is $v = ab$.

This is a mixture of survivor distributions, not a squared norm of an averaged profile. In particular ν_U generally differs from $\|\mu_U\|^2$. Retaining both moments is necessary for the expected distance calculation; replacing ν by the norm of μ would underestimate variance.

Step 3: residual sums above or equal to one. If $q = a + b > 1$, then $\alpha = 1 - a$ and $\beta = 1 - b$. The positive move, of probability $(1 - b)/(2 - q)$, gives $(1, q - 1)$ and leaves the right index. The negative move, of probability $(1 - a)/(2 - q)$, gives $(q - 1, 1)$ and leaves the left index. Hence

$$r_U = q - 1, \quad \lambda = \frac{1 - a}{2 - q}, \quad (38)$$

$$\mu_U = \lambda\mu_L + (1 - \lambda)\mu_R, \quad \nu_U = \lambda\nu_L + (1 - \lambda)\nu_R. \quad (39)$$

The variance coefficient is $v = (1 - a)(1 - b)$. Since $a, b < 1$, $q < 2$ and the denominator is positive. All mixture weights belong to $[0, 1]$.

If $q = 1$, both outcomes are integral: $(1, 0)$ or $(0, 1)$. There is no survivor and its moments are set to zero, but the pivot cost remains present, with $v = ab = (1 - a)(1 - b)$. No formula with denominator zero is used. If exactly one child residual is zero, there is no pivot and the other child's survivor moments pass to the parent unchanged; $A_L + A_R$ is still retained. If both residuals are zero, the parent has no survivor and its cost is again their sum. These cases define the recursion for all possible fractions.

Step 4: close the expected-cost recursion. When both survivors exist, denote their independent profiles by $Z_L = z_{I_L}$ and $Z_R = z_{I_R}$. Independence follows from the fixed conditional tree and independent coins in disjoint subtrees. Expanding their squared distance and taking expectation gives

$$\mathbb{E}\|Z_L - Z_R\|^2 = \mathbb{E}\|Z_L\|^2 + \mathbb{E}\|Z_R\|^2 - 2\mathbb{E}\langle Z_L, Z_R \rangle \quad (40)$$

$$= \nu_L + \nu_R - 2\langle \mu_L, \mu_R \rangle. \quad (41)$$

For the last equality one may expand the finite-dimensional inner product coordinate by coordinate and factor the expectation of each product. The scalar v is deterministic given the child residuals. The expected new pivot cost is consequently v times (41). Adding the already accumulated child costs yields (7).

Linearity suffices to add those costs even though a child's cost and its survivor identity can be dependent. The recursion does not factor such a pair. The only factorization is between distinct subtrees in (41). By induction, every state computed by these equations equals its defining expectation.

At the root, $r_{\text{root}} = \{s_k\}$, and the sole remaining Bernoulli draw has conditional variance h_k at its surviving index. Its expected cost is $h_k \nu_{\text{root},k}$. The martingale covariance identity makes total expected squared empirical error equal to the sum of pivot costs and this terminal cost. Summing columns and restoring Δ^2 establishes

$$\hat{R} = \Delta^2 \sum_k (A_{\text{root},k} + h_k \nu_{\text{root},k}).$$

The normalization $z_i = (a_i^{(1)}, \dots, a_i^{(m)})/\sqrt{m}$ makes its norm exactly the empirical mean square, including the factor $1/m$ in (1).

Step 5: arithmetic, storage, and candidate selection. A full binary tree has V leaves and $V - 1$ internal nodes. For one column, a node needs an inner product of length m , two vector mixtures of the same length at most, and a fixed number of scalar operations. Thus its work is $O(m)$ and the column's work is $O(mV)$; leaf initialization is within this bound. Storing all node means plus scalar moments uses $O(mV)$ auxiliary space. Processing the d columns sequentially gives $O(dmV)$ arithmetic operations without multiplying this auxiliary space by d . The fixed input table and profiles are excluded from the auxiliary count. These are real-arithmetic bounds and do not claim a bound on bit complexity.

For K supplied trees, repeat the evaluation K times for $O(KdmV)$ work. Since K is finite and nonzero, an empirical minimizer exists; choose a deterministic tie-breaking order if needed for measurability. The evaluated risk equals the policy's expected empirical risk, so this selected tree has $\varepsilon = 0$ among the supplied candidates. The event in Theorem 1 holds for every policy even when the candidates were generated from the calibration sample. It therefore gives (5) with the infimum over that realized finite set. The count does not include the cost of constructing the candidates or assert optimality over all trees.

A rational boundary check. For four leaves with fractions $(1/5, 3/10, 3/5, 7/10)$ and scalar profiles $(1/10, 1/5, 3/10, 2/5)$, pair leaves 1, 2 and 3, 4 first. The left residual is $1/2$, $\mu_L = 4/25$, $\nu_L = 7/250$, and $A_L = 3/5000$. The right residual is $3/10$, $\mu_R = 12/35$, $\nu_R = 3/25$, and $A_R = 3/2500$. At the root $q = 4/5$, $v = 3/20$, and the left survivor weight is $5/8$. Hence

$$\nu_{\text{root}} = \frac{5}{8} \frac{7}{250} + \frac{3}{8} \frac{3}{25} = \frac{1}{16}, \quad h = \frac{4}{25}, \quad A_{\text{root}} = \frac{3}{5000} + \frac{3}{2500} + \frac{3}{20} \left(\frac{7}{250} + \frac{3}{25} - 2 \frac{4}{25} \frac{12}{35} \right).$$

Adding $h\nu_{\text{root}}$ gives $307/17500$ at unit grid spacing. This check includes both below-one and above-one child merges and a nonzero terminal contribution.

If every fraction is integral, all costs are zero. If $V = 1$, only the root Bernoulli term remains. For residual sums exactly one, the separate no-survivor branch applies. In numerical computation, testing proximity to these boundaries is a tolerance choice, while the mathematical recursion uses exact equality. The induction, factored moment, operation count, and conditional application of the calibration event together prove Theorem 3. $\square$

B EXPERIMENTAL DETAILS

B.1 DATA AND REPRESENTATION CONSTRUCTION

The text study uses the SetFit JSONL conversion of the public AG News training split, pinned to repository commit `ca5ba619eb034211db5f70932b6702efd21e7c73`. Its source is

https://huggingface.co/datasets/SetFit/ag_news/tree/ca5ba619eb034211db5f70932b6702efd21e7c73. The acquired file contains 33,838,305 bytes and has SHA-256 `ea432c816ef5219d8bd1e049d63f5e05ad76b35e5efd15420d48a96f15ddaa94`. The creator-maintained AG News description states research/noncommercial use; a formal SPDX license was not specified in the inspected material. The study uses the local corpus for research and does not distribute its raw text with the manuscript. Class labels and label text are not used.

We permute all row indices with NumPy’s default generator and seed 20260915. The first 5000 rows fit the table; the next 2048 rows define the evaluation population. The two index sets are disjoint and are preserved with the fitted table. Tokenization lowercases text and extracts maximal matches of `[a-z]+`. There is no stopword removal. The 63 most frequent tokens in the fitting subset receive individual coordinates; every remaining token maps to coordinate 64. A row is normalized by its total token count; a token-empty row is assigned entirely to the other-token coordinate. Thus each profile lies in $\mathcal{S}_{64}$.

Let A_{fit} contain these 5000 normalized histograms and form $G = A_{\text{fit}}^{\top} A_{\text{fit}} / 5000$. Compute its symmetric eigendecomposition, retain the eight eigenvectors with largest eigenvalues, and multiply each retained vector by its eigenvalue to the power $1/4$. Divide the resulting table by its largest absolute entry. This produces an eight-dimensional table in $[-1, 1]$ without using the calibration population or class labels. It is a deliberately small spectral representation of text counts, not a pretrained language-model embedding.

At bit budget b , the 2^b grid points span $[-1, 1]$, so $\Delta = 2/(2^b - 1)$. The lower adjacent index is the floor of $(E + 1)/\Delta$, capped at $2^b - 2$ to allow the upper endpoint; its fraction is clipped to $[0, 1]$ only to absorb floating-point endpoint error. The fitted table itself is not clipped. Nearest rounding uses NumPy’s nearest-integer rule on the fraction. The vocabulary and grid are held fixed while policy selection changes.

B.2 CANDIDATE GENERATION AND RISK EVALUATION

For text-study seed $s \in \{0, \dots, 19\}$, initialize a generator with $20260915 + 2000 + s$. At each calibration size $m = 32, 128, 512$, draw m indices independently and uniformly with replacement from the 2048-row population. Generate four random balanced trees by permuting the 64 leaves, pairing consecutive leaves, and recursively pairing consecutive internal nodes. The fifth tree is Ward linkage on the columns of the calibration histogram matrix. Scaling all profiles by $1/\sqrt{m}$ leaves this linkage’s merge ordering unchanged.

Use the same five candidates at all three bit budgets for that calibration sample. Evaluate their expected empirical risk with Theorem 3 and select the first minimum. The population risk uses the selected tree and all 2048 histograms with uniform weights. This expectation is deterministic conditional on the table and selected tree; uncertainty across seeds comes from calibration and candidate generation. The independent baseline uses the exact marginal-variance expression in Theorem 1. Nearest rounding is a deterministic table evaluated on that same population.

The recursion marks fractions within 10^{-12} of zero or one as integral and treats residual sums within 10^{-12} of one as integral. Negative distances caused by numerical roundoff are floored at zero. A rational four-token check in Appendix A.5 must agree with 307/17500 to absolute tolerance 10^{-12} . Every text configuration also checks the independent and diameter upper bounds with tolerance 10^{-10} . These checks target different parts of the computation: the rational example tests both mixture branches and terminal variance; the inequalities check the global risk scale.

For each seed at $m = 128, b = 3$, sample 100 actual pivotal tables using the selected tree. With population Gram matrix G_{pop} and error table $F = Q - E$, compute each realized risk as $\text{tr}(F^{\top} G_{\text{pop}} F)$. Compare the mean with the recursion’s expectation and divide their difference by the sample standard error. The largest absolute standardized difference is 2.3867. Across 20 comparisons this is a diagnostic, not a formal simultaneous confidence statement.

B.3 SYNTHETIC SAMPLING AND UNCERTAINTY

Uniform-bag seed s uses generator seed $20260915 + s$. At each length, draw 5000 multinomial histograms and 32 independent rounded tables for each method. A pivotal table assigns opposite midpoint errors to each fixed adjacent pair of vocabulary indices; its column sum is exact. Both methods reuse each sampled table across all bags. The reported seed risk averages the bag-by-table squared-error matrix. The exact expressions are stored separately from Monte Carlo estimates.

Rare-block seed s uses $20260915 + 1000 + s$. At each m , draw eight independent hidden bits and a multinomial vector of calibration block counts with probabilities $1/(m+1)$ per block and the remaining probability for the filler. Observed blocks reveal their bit. Unseen blocks receive independent fair guesses. If N_{wrong} guesses are wrong, the selected pivotal policy has exact population risk

$$\frac{N_{\text{wrong}}}{8(m+1)}.$$

Each unseen bit is wrong with probability $1/2$, so the learner’s full expectation equals $G(1-p)^m/[16(m+1)]$, exactly the lower expression in Theorem 5. The uncalibrated guessing reference has expectation $G/[16(m+1)]$. The oracle is zero and is stated in the caption rather than placed on a logarithmic axis.

Every plotted mean averages 20 seed values; the standard error is the sample standard deviation with 19 degrees of freedom divided by $\sqrt{20}$. Analytic references have no sampling band. The grid of experimental settings is fixed in the script, and all configurations are retained. We do not select or suppress seeds according to agreement with a theorem. A finite-seed mean can lie outside one standard error of its expectation, and the bands do not certify a trend simultaneously across settings.

B.4 RESOURCES AND REPRODUCIBILITY

The executable study script is `experiments/study.py`, with SHA-256 `96b1dec2527d7c8d5990c86fc6acfc453dfda53e5ecf7feddb3d9327c6d0e201`. The successful runs use Python 3.10.20, NumPy 2.2.6, SciPy 1.15.3, and Matplotlib 3.10.9 on CPU, with two numerical-library threads and a 2 GB memory limit. Complete text, uniform, and rare studies take approximately 11.09, 4.17, and 1.74 seconds, respectively, in the recorded environment. These times include different workloads and are not throughput comparisons to alternative compression systems. The candidate-evaluation timing reported in the main text excludes Ward and random-tree construction.

From the paper directory, after acquiring the pinned data at the recorded asset location, the three study invocations have the form

```
python experiments/study.py --mode text --seeds 20 --out
experiments/results/reproduced-text
python experiments/study.py --mode uniform --seeds 20 --out
experiments/results/reproduced-uniform
python experiments/study.py --mode rare --seeds 20 --out
experiments/results/reproduced-rare
```

The outputs preserve seed-level rows, calibration indices, expected risks, selected candidate indices, table samples used for the diagnostic, and PDF/PNG figure exports. The fitted table record also preserves the vocabulary and fitting/evaluation row indices. Reproduction requires compatible numerical libraries; eigenvector signs and near-tied clustering operations may differ across implementations, so the stored fitted table is the reference for exact comparisons.

Full-result SHA-256 digests for the text, uniform, and rare JSON files are, respectively,

```
5c6a8621fd4ca7316be47b86f7b027650eae6a6e38b6f33c8a98de31e7a94691
f48be3052012570fb78eb889e645af3863b4d6096f41940d2ccdadc3815f797a
0820141053f1fb87d05de86d4634b2f35cd71f17ab7e7f5a791d194a04a57109
```